15 Shahid is going to use these instructions to make a fizzy drink.

Mix 5 parts of orange juice with 2 parts of lemonade

Shahid thinks that he has 300 m*l* of orange juice and 200 m*l* of lemonade.

(a) If Shahid is correct, what is the greatest amount of fizzy drink he can make?

Orange gives 300 ÷ 5 = 60 ml per part.

Lemonade gives 200 ÷ 2 = 100 ml per part.

1 mark
Find which ingredient runs out first

	Orange	Lemonade	
×60	5	2	×60
	300	120	

1 mark
Use the limiting scale factor 60

300 + 120 = 420 ml of fizzy drink.

1 mark
Correct greatest amount

420 ml

(3)

Shahid has 300 m*l* of orange juice but he only has 160 m*l* of lemonade.

(b) Does this affect the greatest amount of fizzy drink he can make?
Give a reason for your answer.

No. He needs only 120 ml of lemonade for 300 ml of orange.

160 ml is still enough. The orange juice is still the limit.

1 mark
Explain why the maximum remains 420 ml

(1)

(Total for Question 15 is 4 marks)

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